

## SAFETY DATA SHEET

Creation Date 06-Apr-2021

Revision Date 02-Apr-2024

Revision Number 2

### 1. Identification

**Product Name** Titanium(III) sulfate, 0.06 Mol/liter

**Cat No. :** 802585

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Corrosive to metals

Category 1

#### Label Elements

##### **Signal Word**

Warning

##### **Hazard Statements**

May be corrosive to metals

**Precautionary Statements****Prevention**

Keep only in original container

**Spills**

Absorb spillage to prevent material damage

**Storage**

Store in corrosive resistant polypropylene container with a resistant inliner

**Hazards not otherwise classified (HNOC)**WARNING. Cancer - <https://www.p65warnings.ca.gov/>.

### 3. Composition/Information on Ingredients

| Component                              | CAS No     | Weight % |
|----------------------------------------|------------|----------|
| Water                                  | 7732-18-5  | 95       |
| Sulfuric acid                          | 7664-93-9  | 3        |
| Sulfuric acid, titanium(3+) salt (3:2) | 10343-61-0 | 2        |

### 4. First-aid measures

|                                            |                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                      |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable.                                                                                      |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                             |

### 5. Fire-fighting measures

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Unsuitable Extinguishing Media</b>   | No information available |
| <b>Flash Point</b>                      | No information available |
| <b>Method -</b>                         | No information available |
| <b>Autoignition Temperature</b>         | No information available |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |

**Sensitivity to Static Discharge** No information available

### Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition.

### Hazardous Combustion Products

Titanium oxides. Sulfur oxides.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
2

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

### Personal Precautions

Ensure adequate ventilation. Use personal protective equipment as required.

### Environmental Precautions

Should not be released into the environment. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up** Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

### Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Do not get in eyes, on skin, or on clothing.

### Storage.

Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. .

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component     | ACGIH TLV                  | OSHA PEL                                                       | NIOSH                                                  | Mexico OEL (TWA)           |
|---------------|----------------------------|----------------------------------------------------------------|--------------------------------------------------------|----------------------------|
| Sulfuric acid | TWA: 0.2 mg/m <sup>3</sup> | (Vacated) TWA: 1 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | IDLH: 15 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | TWA: 0.2 mg/m <sup>3</sup> |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location.

### Personal Protective Equipment

#### Eye/face Protection

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

#### Respiratory Protection

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

#### Recommended Filter type:

Particulates filter conforming to EN 143.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**9. Physical and chemical properties**

|                                        |                             |
|----------------------------------------|-----------------------------|
| Physical State                         | Liquid                      |
| Appearance                             | Colorless                   |
| Odor                                   | No information available    |
| Odor Threshold                         | No information available    |
| pH                                     | No information available <2 |
| Melting Point/Range                    | No data available           |
| Boiling Point/Range                    | No information available    |
| Flash Point                            | No information available    |
| Evaporation Rate                       | No information available    |
| Flammability (solid,gas)               | Not applicable              |
| Flammability or explosive limits       |                             |
| Upper                                  | No data available           |
| Lower                                  | No data available           |
| Vapor Pressure                         | No information available    |
| Vapor Density                          | No information available    |
| Density                                | 1.15                        |
| Specific Gravity                       | No information available    |
| Solubility                             | miscible                    |
| Partition coefficient; n-octanol/water | No data available           |
| Autoignition Temperature               | No information available    |
| Decomposition Temperature              | No information available    |
| Viscosity                              | No information available    |

**10. Stability and reactivity**

**Reactive Hazard** None known, based on information available

**Stability** Stable under normal conditions.

**Conditions to Avoid** Incompatible products.

**Incompatible Materials**

**Hazardous Decomposition Products** Titanium oxides, Sulfur oxides

**Hazardous Polymerization** Hazardous polymerization does not occur.

**Hazardous Reactions** None under normal processing.

**11. Toxicological information****Acute Toxicity****Product Information**

**Oral LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

**Dermal LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

**Vapor LC50** Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.

**Component Information**

| Component     | LD50 Oral          | LD50 Dermal | LC50 Inhalation               |
|---------------|--------------------|-------------|-------------------------------|
| Water         | -                  | -           | -                             |
| Sulfuric acid | 2140 mg/kg ( Rat ) | Not listed  | LC50 = 0.375 mg/L ( Rat ) 4 h |

**Toxicologically Synergistic** No information available

**Products**

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | No information available                                                                 |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                              | CAS No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------------------------|------------|------------|------------|------------|------------|------------|
| Water                                  | 7732-18-5  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Sulfuric acid                          | 7664-93-9  | Group 1    | Known      | A2         | X          | A2         |
| Sulfuric acid, titanium(3+) salt (3:2) | 10343-61-0 | Not listed | Not listed | Not listed | Not listed | Not listed |

IARC (International Agency for Research on Cancer)

IARC (International Agency for Research on Cancer)

Group 1 - Carcinogenic to Humans

Group 2A - Probably Carcinogenic to Humans

Group 2B - Possibly Carcinogenic to Humans

NTP: (National Toxicity Program)

Known - Known Carcinogen

Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen

ACGIH: (American Conference of Governmental Industrial Hygienists)

A1 - Known Human Carcinogen

A2 - Suspected Human Carcinogen

A3 - Animal Carcinogen

ACGIH: (American Conference of Governmental Industrial Hygienists)

Mexico - Occupational Exposure Limits - Carcinogens

Mexico - Occupational Exposure Limits - Carcinogens

A1 - Confirmed Human Carcinogen

A2 - Suspected Human Carcinogen

A3 - Confirmed Animal Carcinogen

A4 - Not Classifiable as a Human Carcinogen

A5 - Not Suspected as a Human Carcinogen

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component     | Freshwater Algae | Freshwater Fish                                  | Microtox | Water Flea        |
|---------------|------------------|--------------------------------------------------|----------|-------------------|
| Sulfuric acid | -                | LC50: > 500 mg/L, 96h static (Brachydanio rerio) | -        | EC50: 29 mg/L/24h |

**Persistence and Degradability** May persist based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN3264  
 Proper Shipping Name CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S.  
 Technical Name Sulfuric acid  
 Hazard Class 8  
 Packing Group III

#### TDG

UN-No UN3264  
 Proper Shipping Name CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S.  
 Hazard Class 8  
 Packing Group III

#### IATA

UN-No UN3264  
 Proper Shipping Name CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S.  
 Hazard Class 8  
 Packing Group III

#### IMDG/IMO

UN-No UN3264  
 Proper Shipping Name CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S.  
 Hazard Class 8  
 Packing Group III

### 15. Regulatory information

#### United States of America Inventory

| Component                              | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|----------------------------------------|------------|------|-----------------------------------------------|-----------------------------|
| Water                                  | 7732-18-5  | X    | ACTIVE                                        | -                           |
| Sulfuric acid                          | 7664-93-9  | X    | ACTIVE                                        | -                           |
| Sulfuric acid, titanium(3+) salt (3:2) | 10343-61-0 | X    | ACTIVE                                        | -                           |

#### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

#### International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS).

| Component                              | CAS No     | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|----------------------------------------|------------|-----|-----|-----------|-------|------|------|------|-------|----------|
| Water                                  | 7732-18-5  | X   | -   | 231-791-2 | X     | X    |      | X    | X     | KE-35400 |
| Sulfuric acid                          | 7664-93-9  | X   | -   | 231-639-5 | X     | X    | X    | X    | X     | KE-32570 |
| Sulfuric acid, titanium(3+) salt (3:2) | 10343-61-0 | -   | X   | 233-749-9 | -     | -    | X    | X    | X     | KE-12744 |

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component     | CAS No    | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|---------------|-----------|----------|-------------------------------|---------------------------------|
| Sulfuric acid | 7664-93-9 | 3        | 1.0 %                         | -                               |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

| Component     | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|---------------|----------------------------|-----------------------------|------------------------|---------------------------|
| Sulfuric acid | X                          | 1000 lb                     | -                      | -                         |

**Clean Air Act**

Not applicable

**OSHA** - Occupational Safety and Health Administration

Not applicable

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355).

| Component     | Hazardous Substances RQs | CERCLA Extremely Hazardous Substances RQs | SARA Reportable Quantity (RQ) |
|---------------|--------------------------|-------------------------------------------|-------------------------------|
| Sulfuric acid | 1000 lb                  | 1000 lb                                   | 1000 lb<br>454 kg             |

**California Proposition 65**

This product contains the following Proposition 65 chemicals.

| Component     | CAS No    | California Prop. 65 | Prop 65 NSRL | Category   |
|---------------|-----------|---------------------|--------------|------------|
| Sulfuric acid | 7664-93-9 | Carcinogen          | -            | Carcinogen |

**U.S. State Right-to-Know Regulations**

| Component     | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|---------------|---------------|------------|--------------|----------|--------------|
| Water         | -             | -          | X            | -        | -            |
| Sulfuric acid | X             | X          | X            | X        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): Y  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security**

This product does not contain any DHS chemicals.

**Other International Regulations****Mexico - Grade**

No information available

**Authorisation/Restrictions according to EU REACH**

| Component                              | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|----------------------------------------|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Water                                  | 7732-18-5  | -                                                                   | -                                                                             | -                                                                                                     |
| Sulfuric acid                          | 7664-93-9  | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |
| Sulfuric acid, titanium(3+) salt (3:2) | 10343-61-0 | -                                                                   | -                                                                             | -                                                                                                     |

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                              | CAS No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|----------------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Water                                  | 7732-18-5  | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Sulfuric acid                          | 7664-93-9  | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Sulfuric acid, titanium(3+) salt (3:2) | 10343-61-0 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component                              | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Water                                  | 7732-18-5  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Sulfuric acid                          | 7664-93-9  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y34                      |
| Sulfuric acid, titanium(3+) salt (3:2) | 10343-61-0 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By**

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www.thermofisher.com

**Creation Date**

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**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text



**End of SDS**